George Kenison

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Academic Interests: Linear and Algebraic Dynamical Systems, Theory of Computation, Symbolic and Algebraic Computation, Program Analysis and Verification.

Employment

Assistant Professor in Computer Science, **May 2026–**
Department of Computer Science, Maynooth University.

Early Career Researcher, *Banach Center, Institute of Mathematics of the Polish Academy of Sciences*, Simon's Semester on Continued Fractions, Fractals, Ergodic theory, and Dynamics. **May 2026**

Senior Postdoctoral Fellow in Program Verification, Theory, and Algorithms, **February 2026–**
Department of Computer Science, KU Leuven.

Research Fellow in Computer Science and Informatics., **September 2023–September 2025**
School of Computer Science and Mathematics, Liverpool John Moores University.

Postdoctoral Researcher in Automated Reasoning and Program Analysis., **April 2021–August 2023**
Institute of Logic and Computation, TU Wien.

Postdoctoral Researcher in Infinite-State Systems and Dynamical Systems., **June 2018–March 2021**
Department of Computer Science, University of Oxford.

Education

PhD in Mathematics, *University of Warwick*. **2013–2017**
Thesis: Asymptotics in conjugacy classes for free groups. EPSRC doctoral award studentship.

MMath (Masters of Mathematics), *University of Warwick*, *First Class Hons*. **2009–2013**
Dissertation: Periodic orbits of hyperbolic and quasi-hyperbolic toral automorphisms

Peer-Reviewed Conference Papers

- **2024**. **The Threshold Problem for Hypergeometric Sequences with Quadratic Parameters**. In: *51st International Colloquium on Automata, Languages, and Programming (ICALP 2024)*. Vol. 297, 145:1–145:20.
- **2024** (**with** S. Hitarth, L. Kovács, and A. Varonka). **Linear Loop Synthesis for Quadratic Invariants**. In: *41st International Symposium on Theoretical Aspects of Computer Science, STACS 2024*. Vol. 289, 41:1–41:18.
- **2023** (**with** L. Kovács and A. Varonka). **From Polynomial Invariants to Linear Loops**. In: *Proceedings of the 2023 International Symposium on Symbolic and Algebraic Computation, ISSAC 2023*, pp. 398–406.
- **2023** (**with** K. Nosan, M. Shirmohammadi, and J. Worrell). **The Membership Problem for Hypergeometric Sequences with Quadratic Parameters**. In: *Proceedings of the 2023 International Symposium on Symbolic and Algebraic Computation, ISSAC 2023*, pp. 407–416.
- **2023** (**with** J. Nieuwveld, J. Ouaknine, and J. Worrell). **Positivity Problems for Reversible Linear Recurrence Sequences**. In: *50th International Colloquium on Automata, Languages, and Programming, ICALP 2023*, 130:1–130:17.
- **2022**. **On the Skolem Problem for Reversible Sequences**. In: *International Symposium on Mathematical Foundations of Computer Science, MFCS 2022*, 61:1–61:15.
- **2022** (**with** D. Amrollahi, E. Bartocci, L. Kovács, M. Moosbrugger, and M. Stankovič). **Solving Invariant Generation for Unsolvable Loops**. In: *Static Analysis: 29th International Symposium, SAS 2022*. Radhia Cousot Award winning paper, pp. 19–43.
- **2021** (**with** O. Klurman, E. Lefaucheux, F. Luca, P. Moree, J. Ouaknine, M. A. Whiteland, and J. Worrell). **On Positivity and Minimality for Second-Order Holonomic Sequences**. In: *International Symposium on Mathematical Foundations of Computer Science, MFCS 2021*, 67:1–67:15.
- **2020** (**with** R. Lipton, J. Ouaknine, and J. Worrell). **On the Skolem Problem and Prime Powers**. In: *Proceedings of the 45th International Symposium on Symbolic and Algebraic Computation, ISSAC 2020*, pp. 289–296.

Peer-Reviewed Journal Articles

- **2026** (**with** R. Ait El Manssour, M. Shirmohammadi, A. Varonka, and J. Worrell). **Determination Problems for Orbit Closures and Matrix Groups**. In: *Proc. ACM Program. Lang.* 10.POPL. arXiv: `2407.04626 [cs.CC]`.
- **2025** (**with** R. Ait El Manssour, M. Shirmohammadi, and A. Varonka). **Simple Linear Loops: Algebraic Invariants and Applications**. In: *Proc. ACM Program. Lang.* 9.POPL, pp. 745–771.
- **2024** (**with** D. Amrollahi, E. Bartocci, L. Kovács, M. Moosbrugger, and M. Stankovič). **(Un)Solvable loop analysis**. In: *Formal Methods in System Design* SAS 2022.
- **2019** (**with** R. Sharp). **Statistics in conjugacy classes in free groups**. In: *Geometriae Dedicata* 198.1, pp. 57–70.

- **2017** (**with** R. Sharp). **Orbit counting in conjugacy classes for free groups acting on trees**. In: *Journal of Topology and Analysis* 9.4, pp. 631–647.

Preprints

- (**with** O. Klurman, E. Lefaucheux, F. Luca, P. Moree, E. C. Sertöz, J. Ouaknine, A. Whiteland, and J. Worrell). **On the Positivity Problem for second-order holonomic sequences**.
- (**with** P. C. Bell, R. Niskanen, I. Potapov, and P. Semukhin). **On Word Representations and Embeddings in Complex Matrices**.
- (**with** J. Konieczny, F. Luca, A. Scoones, M. Shirmohammadi, and J. Worrell). **On the growth of hypergeometric sequences**. arXiv: `2507.22437 [math.NT]`.

Teaching Experience

Academic Admissions, *University of Oxford*. **2025–2026**
- **Undergraduate admissions lead interviewer** for computer science and joint schools at Wadham College, Oxford.
- **Postgraduate admissions lead interviewer** for the the Department of Computer Science, Oxford.

Co-lecturer for MSc course on *Formal Methods*, **2022, 2023**
Institute of Logic and Computation, Technische Universität Wien.

Co-lecturer for MSc course *Probabilistic Model Checking*, **2019–2020**
Department of Computer Science, University of Oxford.

Stipendiary Lecturer in Pure Mathematics, *St Peter's College, Oxford*. **2018–2020**
- **Academic tutor** for second year undergraduates. Tutorials in *Linear Algebra*, *Integration*, *Group Theory*, and *Graph Theory*. Duties included feedback, assessment, and writing progression reports.
- **Admissions interviewer** for mathematics and joint schools.

Teaching Associate, *School of Mathematics, University of Bristol*. **2017–2018**
- **Academic tutor** for *Linear Algebra*, *Calculus*, *Metric Spaces*, and *Geometry*.

Teaching Assistant, *Mathematics Institute, University of Warwick*. **2013–2017**
- **Undergraduate supervisor**. Small group teaching across the first year mathematics curriculum.
- **Support classes** in *Analysis, Metric Spaces, Experimental Maths, Mathematical Computing*, and *Dynamical Systems*.

Departmental Award for Outstanding Teaching, *Mathematics Institute, University of Warwick*.

Student Feedback.
- "Always prepared, always cheerful and always willing to go that extra mile in helping students to understand—a true inspiration!"
- "He was engaging, whilst provoking the students to find their own way to the answers."
- "I was involved in an incident in term one and if it wasn't for his support, both [academic and pastoral], I wouldn't have made it through the term and hence the year."
- "George made me feel comfortable asking questions and. . . his analysis classes were a highlight of my week."

Selected Seminar Talks

Decision Problems for P-finite Sequences, *Warsaw*. **July 2025**

The Threshold Problem for Hypergeometric Sequences with Quadratic Parameters, *Liverpool*. **June 2024**

From Polynomial Invariants to Linear Loops, *IRIF, Paris*. **Dec 2023**

(Un)Solvable Loop Analysis, *Liverpool*. **Oct 2023**

Decision Problems for Hypergeometric Sequences, *TU Wien and ISTA*. **Feb 2023**

On the Skolem Problem and Reversible Sequences, *Chalmers and Gothenburg*. **July 2022**

On the Skolem Problem and Reversible Sequences, *TU Wien and ISTA*. **May 2022**

On Positivity and Minimality for Second-Order Holonomic Sequences, *Open University*. **Sept 2021**

On Positivity and Minimality for Second-Order Holonomic Sequences, *TU Wien and ISTA*. **Sept 2021**

Skolem's Problem and prime powers, *Bristol*. **Dec 2018**

Selected Conference and Workshop Talks

Determination Problems for Orbit Closures and Matrix Groups, *Rennes*, **Jan 2026**
Symposium on Principles of Programming Languages.

The Membership Problem for Hypergeometric Sequences, *Warsaw*, **Jun 2025**
Workshop on Series, Automata, Matrices, Symbolic dynamics, and their Applications.

The Threshold Problem for Hypergeometric Sequences with Quadratic Parameters, *Vienna*, **Sept 2024**
International Conference on Reachability Problems.

The Threshold Problem for Hypergeometric Sequences with Quadratic Parameters, *Tallinn*, International Colloquium on Automata, Languages, and Programming. **July 2024**

The Membership Problem for Hypergeometric Sequences with Quadratic Parameters, *Tromsø*, International Symposium on Symbolic and Algebraic Computation. **July 2023**

On the Skolem Problem for Reversible Sequences, *Kaiserslautern*, International Conference on Reachability Problems. **Oct 2022**

On the Skolem Problem for Reversible Sequences, *Vienna*, International Symposium on the Mathematical Foundations of Computer Science. **Aug 2022**

On Positivity and Minimality for Second-Order Holonomic Sequences, *Tallinn*, International Symposium on the Mathematical Foundations of Computer Science. **Aug 2021**

On the Skolem Problem and Prime Powers, *Kalamata*, International Symposium on Symbolic and Algebraic Computation. **July 2020**

Widening Participation and Outreach

Mathematics in Education and Industry, *Problem Solving Matters: tutor and mentor*. **Summer 2017**

Further Maths Support Programme, *STEP/AEA workshops, enrichment days, Royal Institution masterclasses, and problem solving classes*. **2012–2019**

Academic Service

Program Committee Member.

- RP. *International Conference on Reachability Problems 2025, 2024*
- SCSC. *International Workshop on Satisfiability Checking and Symbolic Computation 2026*

External Reviewer/Sub-Reviewer (Conferences).

- CASC. *Computer Algebra in Scientific Computing 2022*
- CONCUR. *International Conference on Concurrency Theory 2023*
- ICALP. *International Colloquium on Automata, Languages, and Programming 2026, 2025, 2023, 2020*
- ISSAC. *International Symposium on Symbolic and Algebraic Computation 2026, 2024*
- LICS. *Symposium on Logic in Computer Science 2026, 2025*
- MFCS. *International Symposium on Mathematical Foundations of Computer Science 2023, 2021*
- POPL. *Principles of Programming Languages 2023*
- SODA. *Symposium on Discrete Algorithms 2024*
- STACS. *Symposium on Theoretical Aspects of Computer Science 2025, 2023*
- STOC. *Symposium on Theory of Computing 2025, 2024*
- TACAS. *Conference on Tools and Algorithms for the Construction and Analysis of Systems 2023*

External Reviewer/Sub-Reviewer (Journals).

- Ex. Countex. *Examples and Counterexamples*
- FI. *Fundamenta Informaticae*
- Int. J. Math. Comput. Sci. *International Journal of Mathematics and Computer Science*
- J. ACM. *Journal of the Association of Computing Machinery*
- J. Symbolic Comput. *The Journal of Symbolic Computation*
- MCS. *Mathematics in Computer Science*
- TOPLAS. *ACM Transactions on Programming Languages and Systems*

Workshop Organiser.

- *Workshop on Loop Invariants and Algebraic Reasoning*, ICALP 2025 Satellite Workshop.
- *Workshop on Reachability, Recurrences, and Loops*, ICALP 2023 Satellite Workshop.

Autobóz, *Annual Research Camp on Automata, Logic, and Games*.

- **Steering Committee,** 2023–
- **Organiser,** Autobóz 2023 (16–22 July) in partnership with the *Highlights Collaborative Research Week*.
- **Speaker** at Autobóz 2023 (Tutorials on Decision Problems for Linear Recurrence Sequences).

Departmental & University Service

Pay and remuneration committee for sessional teachers, *Warwick*. **2015–2017**

Staff & Graduate Student Liaison Committee (Mathematics), *Warwick*. **2014–2017**

Professional Development

Academic Leadership Development Course, *Informatics Europe*.

Effective Research Project Proposal Writing for Public Funding, *TU Wien*.

Cultivating Research-rich Education and Training Excellence (FHEA certified), *University of Bristol*.

Postgraduate Award in Teaching and Learning in Higher Education (AFHEA certified), *University of Warwick*.